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An Ensemble Machine Learning-Based Method to Determine the Presence of Heart Ailment Using Pooled Dataset

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***Abstract*— Heart disease remains the leading cause of mortality globally, emphasizing the critical need for better diagnostics. This study used a variety of machine learning methods to forecast cardiac disease, using SVM, Random Forest, Decision Tree, KNN, and LGBM. The evaluation measures include accuracy, ROC, precision, recall, and F1 score. Data pretreatment procedures such as cleansing, transformation, integration, and reduction were used to guarantee reliable analysis. The model's performance was further enhanced by hyperparameter tuning. SVM had the lowest performance of the models examined, with 57.23% accuracy, 86% precision, 89% recall, and an F1 score of 87%. The study used three different datasets, combining their strengths to improve model discrimination, decrease overfitting, and increase overall accuracy. We developed multiple models from the combination of RF, DT and LGBM. We found accuracy of up to 99.54% by merging LGBM and Random Forest. These findings illustrate the efficacy of the ensemble approaches and data integration in improving heart disease prediction skills using machine learning.**

Keywords— Machine learning, Heart disease, Random forest, Combined dataset, Ensemble

I. Introduction

Cardiovascular disease (CVD) accounts for more than 70% of all deaths globally and is the main cause of illness and death. According to the 2017 Global Burden of Disease study [1, 2], heart disease accounts for around 43% of the worldwide deaths. According to the Australian Institute of Health and Welfare (AIHW), CVD will account for 42% of all deaths in Australia in 2018 [3]. Numerous factors, such as high blood pressure, obesity, high cholesterol, smoking, drinking, a lack of exercise, diabetes, and irregular cardiac rhythms, may give rise to heart disease but these factors can be controlled by people. Clinical research reveals such as weight gain, a diet high in fat, a low amount of good HDL cholesterol, A substantial amount of LDL (bad) cholesterol with high blood pressure, and a family record of heart disease [4].

Moreover, some consumers find certain technologies—like CT scans and electrocardiograms—too expensive and unfeasible for identifying coronary cardiovascular disease. 17 million individuals have died as a result of the previously mentioned cause alone [5]. According to figures by the World Health Organization (WHO), about 23.6 million deaths worldwide could be attributed to cardiovascular illnesses, primarily strokes and cardiac arrests, by 2030 [6]. As a result, using machine learning and data mining methods to forecast the probability of getting heart disease is critical for saving lives and reducing the financial burden on society. In the medical profession, machine learning is essential. Machine learning enables us to detect, the right path, and estimate an extensive range of ailments. Predicting the likelihood of getting certain diseases using data mining and machine learning techniques has lately gained prominence. The use of data mining methods for illness prediction has already been documented. While some studies have attempted to estimate the likelihood that the condition will worsen in the future, accurate results have yet to be discovered [7].

This paper describes an ensemble model that improves heart disease prediction by combining feature extraction and data balance across three datasets. The following important work are presented in this paper:

- Hyper parameter tuning is used for optimizing the hyper parameters of machine learning models to improve its performance and null value handling, feature scaling, data balancing, missing data handling are used for data preprocessing.
- Several machine learning classifiers such as random forest (RF), k-nearest neighbor (KNN), Light gradient boosting machine(LGBM), support vector machine (SVM) and Decision Tree (DT) are utilized in this paper.
- To improve the model's accuracy, models that achieved highest accuracy were finally integrated using the soft voting ensemble method.

II. Literature REview

The machine learning (ML) algorithms were extensively studied for predicting cardiovascular disease (CVD) based on clinical data. These datasets provide issues because to their

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complexity and class imbalance. Without addressing these challenges, ML models may suffer from decreased efficiency and accuracy. Prior research has used a variety of feature selection approaches and machine-learning techniques to address these issues.

Uddin et al. [8] trained eleven machine learning algorithms, including SVM, Random Forest, and Decision Tree, using a dataset compiled from UCI, IEEE Data Port, and Kaggle to detect heart disease. They attained an incredible 99.16% accuracy using the Decision Tree. However, such high accuracy may suggest overfitting, in which the model performs well on training data but badly on unknown data.

Javed et al. [9] sought to improve classification accuracy by combining Huang's technique with k-mode clustering and employing systems including XGBoost, DT, RF, and Multilayer Perceptron. They used GridSearchCV for hyperparameter tweaking. MLP emerged as the strongest model, particularly after cross-validation, with AUCs above 0.9. The concern with this strategy is that minor accuracy differences with and without cross-validation indicate that the model's performance may be unstable, possibly compromising its dependability.

Chandrasekhar [10] built a diagnostic approach that utilized five-fold cross-validation and GridSearchCV for optimization. They tested six classifiers, and Logistic Regression and AdaBoost had had an accuracy rate of 90.16% and 89.67%, correspondingly. The soft voting ensemble classifier achieved up to 95% accuracy. Despite the great accuracy, the findings may have been improved by verifying the models on a larger and more diversified dataset. Before using k-mode clustering to a heart disease dataset, Bhatt et al. [11] converted ages to year bins and separated blood pressure data into intervals. MLP had the highest accuracy, 87.23%. In this scenario, the study's preprocessing stages and clustering strategy resulted in accuracy shifts indicating that the model may fail to generalize across diverse datasets.

Ledig et al. [12] developed a decision assistance tool that includes Hyperband for hyperparameter optimization, extremely randomized trees for classification, and SMOTE for data balancing. Their model obtained 99.2% accuracy on the Cleveland dataset and 98.52% on the Statlog dataset. The disadvantage of this study is that the extremely high accuracy raises worries regarding potential overfitting. Furthermore, while tree-based models performed well, their applicability to different datasets and real-world contexts required additional validation. These studies show the promise of ML in CVD prediction, but they also emphasize the significance of tackling overfitting and verifying models on a variety of datasets to ensure robust and dependable results.

A more extensive and diverse dataset offers a more varied range of instances from which models can be trained. Furthermore, most prior research employed just one heart disease dataset to conduct this experiment. In this study, we employed an innovative mixed heart disease dataset which includes three distinct datasets, and we suggested soft voting methods based on improving several machine learning classifiers: DT, RF and LGBM. With ensemble machine learning approach, merging RF and LGBM classifier the result is increased.

III. Material and Methods

The primary objective of this approach is to estimate an individual's likelihood of having cardiac difficulties using medical data (age, blood pressure, cholesterol, etc.), making it possible to identify the problem quickly and accurately. In this work, five machine learning algorithms were used: SVM, RF, DT, KNN, and LGBM. To predict cardiac disease, the suggested study evaluated a few classification methods and analyzes how they perform. The examination gives an accurate assessment of the individual's heart state. The physician enters the information from the individual's healthcare record. The chance of developing heart disease is predicted using an approach built with information as the source. The entire process is as seen in Fig.1.

A. Datasets

Three datasets were used in this research to project heart disease. These three datasets have been gathered. One from IEEE Data Port [13] and two datasets from Kaggle. [14-15]. The gathered dataset contains 14 characteristics, 13 attribute values, and one targeting variable, although the sample size for this study is 5450. TABLE I displays the overall amount of entries across three distinct data sets as well as the number of characteristics with attribute names.

TABLE I. Datasets for heart disease.

Sr No.	Dataset Name	Attributes		Total Entries
1	Heart Disease Dataset (Comprehensive), IEEE Data port	14	Age, sex, Cp, Trestbps,chol, fbs, restecg, thalach, exang, oldpeak, slope, ca, thal, target	1025
2	Heart Disease Predication, Kaggle	12	Age, sex, Cp, Trestbps,chol, fbs, restecg, thalach, exang, oldpeak, slope, target	3235
3	Heart Disease Predication Dataset, Kaggle	12	Age, sex, Cp, Trestbps,chol, fbs, restecg, thalach, exang, oldpeak, sl_slope, target	1190

The three datasets used unique variables to determine heart disease. It contains the patient's age, gender (0 female, 1 male), and chest pain kind (0 typical angina, 1 uncommon angina, 2 non-anginal discomfort, and 3 asymptomatic). It also records standing blood pressure levels (in mmHg), serum cholesterol (in mg/dl), fasting blood sugar levels over 120 mg/dl (1 for true, 0 for false), and resting ECG readings (0 for normal, 1 for ST-T wave abnormalities, and 2 for left cardiac hypertrophy). Other characteristics include maximum heart rate reached, exercise-induced angina (0 for no, 1 for yes), exercise-induced

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ST depression, slope of the peak exercise ST segment (0 inclining, 1 level, 2 sloping), number of main vessels (0-3), thalassemia (3 normal, 6 fixed defect, 7 reversible defect), and the existence of heart disease (0 no, 1 yes)

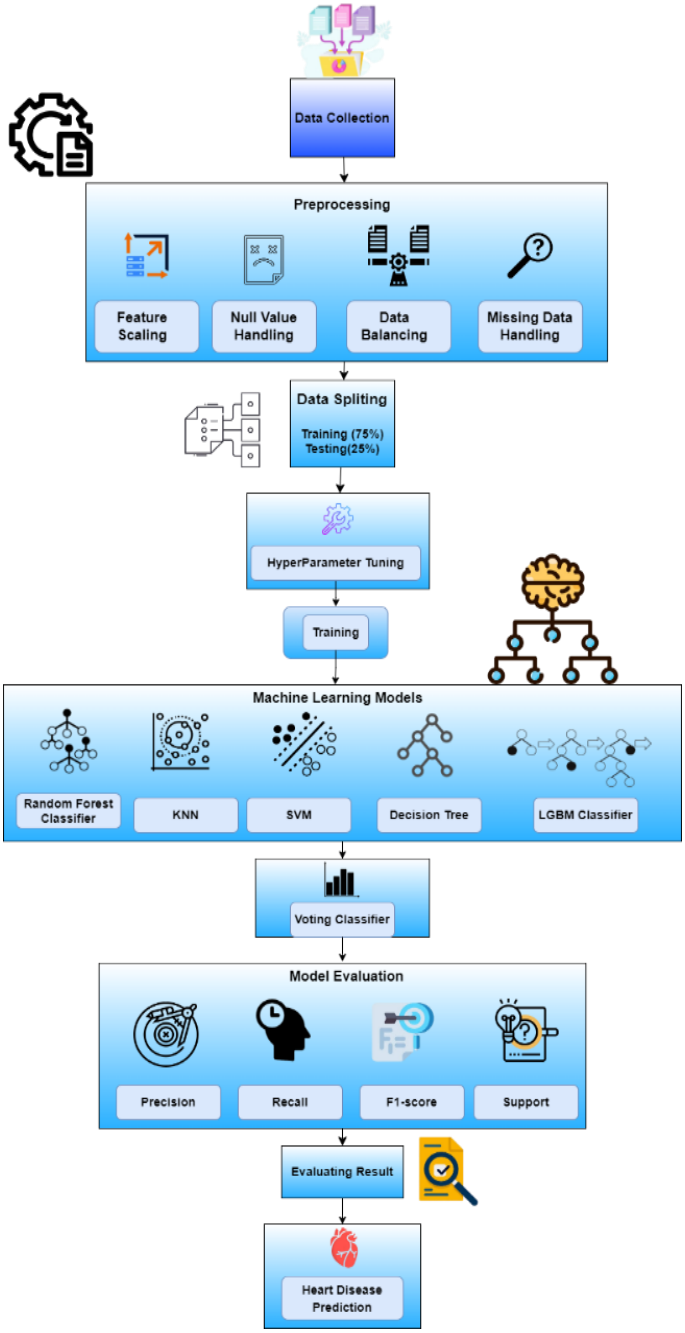


Fig-1. Heart disease prediction model.

B. *Data pre-processing*

In our research, the data preprocessing phase is critical for ensuring the dataset's integrity and quality before feeding it into machine learning models. Here, we discuss the exact processes we took to preprocess the data for predicting heart disease. We utilized Grid Search with Cross-Validation (CV) to examine a variety of Hyperparameters for each model. In the case of RF the number of estimator trees is 100, the maximum tree depth is 20, and the smallest sample size per leaf is 5. The highest cross-validation score for RF is 0.9354 and for LGBM is 0.9482. Similarly, for LGBM the learning rate is 0.1, there are 200 estimator trees with 31 leaves, and the boosting type is GBDT.

Feature scaling- It is a crucial step in the preprocessing phase. Age, blood pressure, and cholesterol levels were all represented in our dataset using different units and scales. To guarantee that these variations would not distort the model's learning process, we normalized the features. By converting each feature to a standard scale, we guaranteed equal contribution for the model's predictions.

Handling Null Value- Data preparation requires careful handling of missing values. Certain data points in our dataset were missing, which might result in erroneous predictions. We resolved this issue with imputation, which replaces missing values with the median values of the corresponding attributes. This method helps to keep the data consistent while avoiding biases.

Data Balancing- Imbalanced datasets can bias models towards the majority, resulting in poor performance for minority classes. Our dataset showed such an imbalance, with fewer incidences of heart disease than non-heart illness patients. To address this, we used the Synthetic Minority Over-sampling Technique, which creates manufactured sample from the minority class, balancing the dataset. This guarantees that the model learns evenly from both groups.

Handling Missing Data- Missing data in medical datasets might result from missing patient records or data entry mistakes. To overcome this, we used median imputation approach. Median imputation was used to replace missing information in numerical variables such as age, cholesterol levels, and blood pressure. The median is less susceptible to outliers than the mean, making it a more reliable choice for dealing with skewed distributions common in medical data.

Data Splitting- Dividing the information into training and testing sets was crucial for our methodology. We used 75% data to train on and kept 25% for assessment.

Feature Scaling: Because the dataset included features in various units (e.g., age in years, cholesterol in mg/dl), feature scaling was required to guarantee that all characteristics contributed equally to the model's learning process. Standardization was selected over normalization because it converts the data to a mean of zero and a standard deviation of one, which is especially beneficial for feature-sensitive models.

These preparation techniques were used to maintain data quality and improve model performance. Median and mode imputation were chosen for their ability to handle outliers, which are prevalent in medical datasets. Feature scaling through standardization improves convergence speed and model performance.

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C. Machine learning Algorithms

Random forest

Random-forest techniques are used for classification and regression analysis. Data is represented as a tree for forecasting purposes [16]. Random Forest may still produce accurate results on huge datasets, even if some record entries are absent. The decision tree's analysis and study may be preserved and utilized in several data sets. A random forest has two stages: first, it generates the random forest, and then it projects the categorization established in the first phase. Equation (1) uses pi to calculate the chance of an item being categorized as a specific class or characteristic.

$$\text{Gini}=1-\sum_{i-1}^{n}(pi)^2 \quad (1)$$

The selection criteria focused on achieving a balance between accuracy, interpretability, computational efficiency, and robustness against overfitting.

IV. Result and Discussion

Data was collected, separated between training and evaluation sets, then randomly infused. To avoid input losses and excessive fitting, certain input-prepping methods were solely used during the validation period before being applied to the test sets. The entire process occurred out remotely on the personal computer using Google Co-Lab and Python libraries including Scikit-Learn, NumPy, Pandas, Seaborn, and Matplotlib. TABLE II shows critical components and system requirements for the proposed model.

TABLE II. System specification of the proposed model

System requirements for ML algorithm development	
RAM	12.68 GB
CPU	2- core xeon 2.2 GHz
Cache	56320 KB
GPU	Tesla K80 GPU
GPU Memory	12 GB

This section displays the outcomes of several machine learning methods, such as SVM, DT, RF, AND LGBM and KNN. Equations (2-5) show measures used to evaluate algorithm performance, such as accuracy rate, precision (P), recall (R) and F1-score [17].

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \quad (2)$$

$$Precision = \frac{TP}{TP + FP} \quad (3)$$

$$Recall = \frac{TP}{TP + FN} \quad (4)$$

$$F1\ Score = 2\frac{Precision.Recall}{Precision + Recall} \quad (5)$$

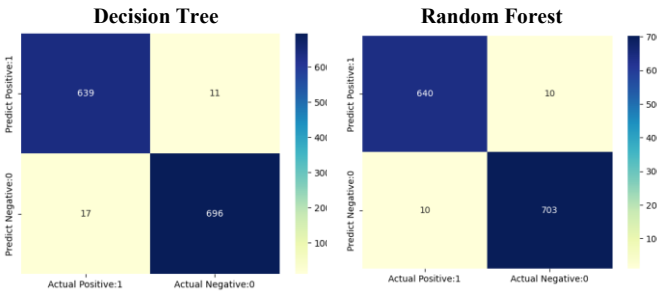
A confusion matrix is a chart that is utilized to assess the efficiency of supervised teaching techniques. The suggested hybrid model, which included RF and LGBM classifiers, had a maximum accuracy of 99.54%, outperforming the baseline models by 1-2%. TABLE III categorizes True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN), all of which are vital to assessing predictive model accuracy, notably in healthcare diagnostics.

TABLE III. Factors comprising the confusion matrix.

Actual Values			
	Negative(0)	**Positive (1)**	
Negative(0)	**TN**	**FN**	**Predicted**
Positive (1)	**FN**	**TN**	

Fig- 2 depicts the confusion matrices for several classifiers, which include the counts of TP, TN, FP and FN. The RF and LGBM classifiers, when combined with other classifiers, produce larger counts of true positives and true negatives. For example, the Random Forest classifier has 640 TP, 703 TN, 10 FP, and 17 FN, but the LGBM classifier has 497 TP, 554 TN, 25 FP, and 14 FN. The confusion matrix for the K Neighbors Classifier (KNN) shows inferior performance (447 TP and 506 TN), suggesting more classification mistakes.

Researchers have explored machine-learning approaches heart disease over several years, with varying levels of effectiveness. Early detection of cardiac disorders is crucial for rehabilitation, as they are becoming increasingly prevalent worldwide. This inspires individuals to accept and overcome the limitations of illness. The analysis of different ML classifiers is shown in TABLE IV.



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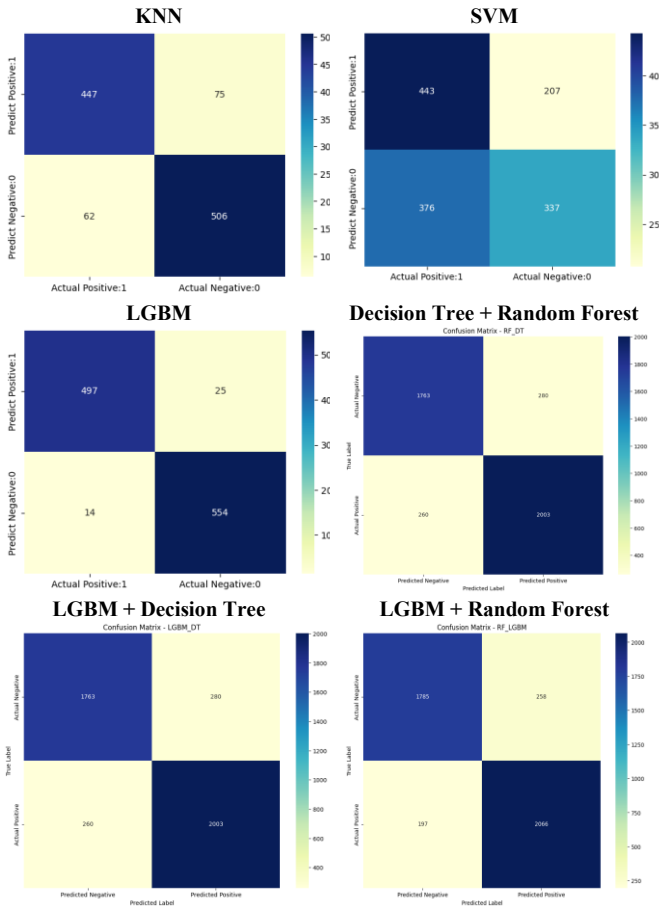


Fig-2. Confusion matrices for several ML classifiers.

TABLE IV. ANALYSIS OF MACHINE LEARNING ALGORITHMS AND THE RESULTS.

Model	Accuracy	Precision	Recall	F1 Score
DT	98.01%	99%	96%	97%
RF	98.60%	99%	98%	99%
KNN	87.01%	88%	90%	89%
SVM	57.22%	85%	89%	87%
LGBM	96.55%	97%	97%	97%
DT+RF	99.36%	99%	96%	97%
LGBM+DT	98.53%	99%	96%	97%
LGBM+RF	99.54%	98%	98%	98%

D. Comparison with existing work

TABLE V shows a comprehensive examination of many machine learning models for predicting outcomes, including parameters like instance accuracy (Acc.), ROC AUC, precision (Prec.), recall, and F1 score. The DT and RF models perform well, with accuracies of 98.01% and 98.60%, high ROC AUC, precision values of 99%, recall rates of 96% and 98%, and F1 scores of 97% and 98%. The KNN model performs well, with 87.01% accuracy and 95% ROC AUC, but it falls behind in other aspects. Despite a high ROC AUC of 97%, the SVM has a lower accuracy of 57.22%, revealing metric inconsistencies. LGBM reaches 96.55% accuracy. Combined models (DT+RF, LGBM+DT, and LGBM+RF) improve performance even further, with LGBM+RF obtaining the best accuracy 99.54%.

TABLE V. COMPARISON WITH PREVIOUS WORKS ON CARDIOVASCULAR DISEASE FORECASTING, UTILIZING MACHINE LEARNING.

Author	Dataset	Method	Accuracy
Maini et al.[19]	Cleveland Heart Disease	ANN	90.74%
Uddin et al. [8]	UCI+IEEE Data port + Kaggle	Decision Tree	99.16%
Bharti et al. [18]	UCI	Deep Learning	94.2%
Chandrasekhar et al. [10]	Cleveland	Soft voting ensemble method	93.44%
Proposed method	Kaggle+Kaggle+IEEE	Hybrid Model	99.54%

Fig-3 represents the ROC curve for testing several classifiers. The RF classifier, LGBM + RF, and DT + RF have the greatest AUC values 99%, indicating greater performance. Other classifiers, for example the DT classifier and the LGBM classifier, are likewise effective, with AUC values of 0.95 and 0.98, respectively. ROC curves (Fig- 3) show that the ensemble models attained an AUC of 99%, exceeding prior research by K.M. Mohi Uddin et al. (2023) and Maini et al. (2021), Who disclosed AUCs of 98% and 90% respectively.

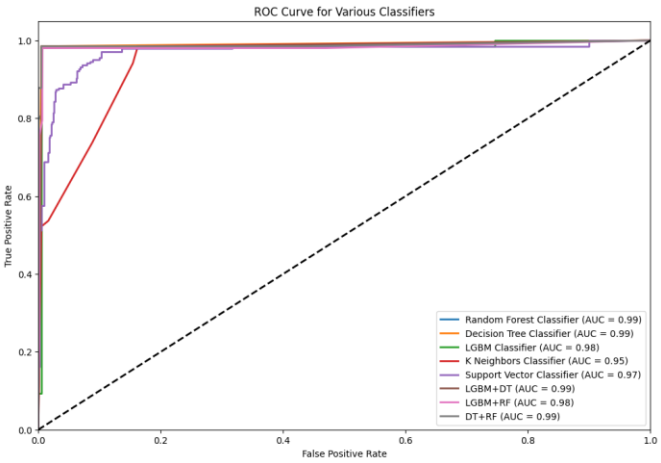


Fig-3 ROC curve for testing various classifiers

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V. Conclusion

Cardiovascular illness happens to be one of the top reasons for mortality around the world. Early detection can stop the illness from being worse. Our aim is to achieve effective prediction while using the combined dataset. Also soft voting classifier was applied to the combined dataset to achieve superior accuracy. In this paper, To train and test the created system, three datasets are integrated and various preprocessing methods like (cleaning, transforming, integrating, and reducing) have been used. We conducted experiments using the following classification models : SVM, Random Forest, Decision Tree, KNN and LGBM. Indicating superior performance in predicting heart disease on the combined dataset, the DT and RF classifiers achieved the maximum accuracies of 98.16% and 98.67%, respectively. Although SVM is a best classifier for predicting heart disease, it performs very poorly on our combined dataset with an accuracy of 57.23%.According to the results, the soft voting ensemble classifier approach produced even higher accuracy of 99.54% when it was applied in Decision Tree + Random Forest, LGBM + Random Forest and Decision Tree + LGBM ,machine learning classifier. This study may be extended to establish the accuracy, fairness, transparency, and results of the model to identify cardiac problems using an explainable machine learning approach. This study can also be enhanced with the use of a self-created dataset.

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